

RMO– 2025

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1.

(a) Let $n \geq 3$ be an integer. Find a configuration of n lines in the plane which has exactly

- (i) $n - 1$ distinct points of intersections.
- (ii) n distinct points of intersections.

(b) Give configurations of n lines that have exactly $n+1$ distinct points intersection for:

- (i) $n = 8$
- (ii) $n = 9$.

2. Let a, b, c be distinct nonzero real numbers satisfying $a + \frac{2}{b} = b + \frac{2}{c} = c + \frac{2}{a}$.

Determine the value of $|a^2b + b^2c + c^2a|$.

3. Let Ω and Γ be circle centered at O_1, O_2 respectively. Suppose that they intersect in distinct points A, B . Suppose O_1 is outside Γ and O_2 is outside Ω . Let ℓ be a line not passing through A and B that intersects Ω at P, R and Γ at Q, S so that P, Q, R, S lie on the line in this order. Furthermore, the points O_1, B lie on one side of ℓ and the points O_2, A lie on the other side of ℓ . Given that the points A, P, Q, O_1 are concyclic and B, R, S, O_2 are concyclic as well. Prove that $AQ = BR$.

4. Prove that there do not exist positive rational numbers x and y that:

$$x + y + \frac{1}{x} + \frac{1}{y} = 2025.$$

5. Let ABC be an acute angled triangle with $AB < AC$, orthocenter H and circumcircle Ω . Let M be the mid point of minor arc BC of Ω . Suppose that MH is equal to the radius of Ω . Prove that $\angle BAC = 60^\circ$.

6. Let $p(x)$ be a nonconstant polynomial with integer coefficients and let $n \geq 2$ be an integer such that no term of the sequence:

$$p(0), p(p(0)), p(p(p(0))), \dots$$

is divisible by n . Show that there exist integers a, b such that $0 \leq a < b \leq n - 1$ and n divides $p(b) - p(a)$.

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